Sections: B,C & D

Linear Algebra (MT-1004)

Homework#26

Instructor: Muhammad Ali

Simple Linear Regression

Question 1

A car dealership wants to predict the selling price of used cars based on their mileage (in kilometers). The dataset is as follows:

Mileage (X in kilometers)	Price (Y in dollars)
35,000	18,500
40,000	17,500
45,000	16,500
50,000	15,800
60,000	14,000

- Write the design matrix for given data
- Compute the coefficients b (Bias) and w_1 (Weights)
- Find the equation of linear regression model on the given data
- Based on these coefficients, predict the selling price of a car with 55,000 kilometers mileage